

Project Design Phase-I

Problem – Solution Fit

Date	16 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	2 Marks

Problem – Solution Fit

Heart Disease Analysis: Transforming Clinical Data into Actionable Insights

Identifying a real problem faced by healthcare professionals and ensuring that the developed solution effectively addresses it is central to this project. In the context of heart disease management, stakeholders such as cardiologists, hospital administrators, and public health researchers struggle to interpret complex and scattered patient datasets.

The Heart Disease Analysis Dashboard transforms raw clinical data into clear, interactive visual insights, enabling accurate diagnosis support, early risk detection, and informed medical decision-making.

Purpose

- **Solve complex healthcare data challenges**

The project converts unstructured patient data (age, cholesterol, blood pressure, heart rate, lifestyle factors, etc.) into interactive dashboards. This helps healthcare professionals easily identify high-risk patients, demographic patterns, and key contributing risk factors.

- **Improve adoption through familiar tools and intuitive design**

By using Tableau or Power BI dashboards accessible via web platforms, medical staff can explore data through simple filters, charts, and drill-down features without requiring advanced technical or programming skills.

- **Strengthen clinical communication and decision-making**

Clear visualizations support effective reporting during clinical meetings and strategy discussions. Doctors and administrators can present patient risk trends, demographic comparisons, and health indicators confidently, improving treatment planning and preventive care strategies.

- **Increase engagement and trust in data-driven healthcare**

By addressing recurring challenges such as difficulty identifying high-risk groups, scattered patient records, and unclear trend patterns, the solution builds confidence in data-driven medical analysis and encourages continuous dashboard use.

• Understand and improve heart disease management

The project provides insights into demographic trends, lifestyle correlations, and clinical indicators influencing heart disease risk. It enables comparison across age groups, gender, and risk categories, helping healthcare stakeholders optimize prevention programs and improve patient outcomes.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Health-conscious individuals, patients with family history, medical professionals, researchers	6. CUSTOMER CONSTRAINTS - Limited medical knowledge - high cost of specialized tests - data privacy concerns - Complex medical reports	5. AVAILABLE SOLUTIONS - Regular check-ups, fitness trackers - online symptom checkers - general health advice websites, - traditional medical consultations	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS - Predict heart disease risk early - Understand personal risk factors - monitor cardiovascular health continuously, receive personalized health insights - Reduce anxiety about heart health	9. PROBLEM ROOT CAUSE - Lack of continuous, integrated health data monitoring - difficulty interpreting complex medical data, delayed detection of subtle symptoms	7. BEHAVIOUR - Regular exercise, dietary tracking - visiting doctors for check-ups - researching health information online, discussing health with family	
Identity strong TR & EM	3. TRIGGERS - Experiencing unusual symptoms (e.g., chest pain), family member diagnosed with heart disease, doctor's warning during a check-up	10. YOUR SOLUTION An AI-driven analysis platform that processes health data from wearables and medical records to provide personalized, real-time heart disease risk assessments and actionable health insights	8. CHANNELS OF BEHAVIOUR 8.1 ONLINE - Health portals, wearable device apps, telemedicine platforms, online health communities	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER Before: Anxious, worried, uncertain, stressed After: Reassured, informed, proactive, confident		8.2 OFFLINE Doctor's clinics, hospitals, health seminars, community health programs	